SAMIR SALMAN

ROME, ITALY

Email: samirsalman.dev@gmail.com **Tel:** +39 3293346168 **Website:** https://samirsalman.github.io/web-site/
Github: https://github.com/samirsalman **Linkedin:** https://www.linkedin.com/in/samir-salman-9bb92112a

MLOPS ENGINEER

MLOps Engineer specializing in NLP and AI infrastructure, with expertise in deploying scalable, high-performance systems and automating end-to-end ML pipelines. Skilled in Kubernetes, Docker, and distributed computing, ensuring reliability and efficiency from research to production. Experienced in Neural Machine Translation, Text-to-Speech, Retrieval-Augmented Generation, and multi-agent AI systems, bridging applied research and engineering to deliver production-ready, intelligent solutions

---

EXPERIENCE

LEAD ML ENGINEER

Stip AI

Rome, Italy | Sep 2024 - present

- Built and led the AI team from the ground up to develop and enhance customer service AI agents, driving innovation in voice and multi-agent systems.
- Defined and implemented the product AI infrastructure, optimizing reliability, availability, and service latency across distributed environments.
- Designed and deployed a multi-agent framework on top of LangGraph and LangChain to support multi-tenant scenarios and reduce response latency.
- Developed and optimized phone-based voice agents to improve conversational naturalness and responsiveness for real-time customer interactions.
- Enhanced Retrieval-Augmented Generation (RAG) pipelines through metadata-enriched information and scalable data ingestion infrastructure.
- Led end-to-end project definition, team management, and CI/CD implementation to ensure seamless model deployment and continuous improvement of AI systems.

MLOPS ENGINEER - AI PRODUCTION TEAM LEAD

Translated

Rome, Italy | Feb 2022 - Sep 2024

- Designed, deployed, and migrated all AI infrastructure to an on-premises, scalable, and geographically distributed Kubernetes cluster, ensuring seamless and reliable AI model deployment.
- Built a lightweight SLURM-based framework to optimize distributed GPU training, cutting cloud idle time by 30% and maximizing resource utilization.
- Led a team of two engineers to develop high-performance libraries, optimizing inference for Neural Machine Translation and Text-to-Speech models, automating pipelines, and enhancing model reliability and accuracy.
- Managed the training and deployment of 100+ Machine Translation models, boosting the average COMET score by 15% across all language pairs.
- Developed a scalable AI pipeline handling terabytes of data, streamlining model training in a multi-gpu and multi-node setting, deployment, and NVIDIA DGX Cloud utilization for efficient production scaling.
- Led the development of an internal AI data annotation tool, generating 100,000+ high-quality human-labeled data points critical for model evaluation and fine-tuning proprietary LLMs.

AI SPECIALIST

Stip

Rome, Italy | Sep 2021 - Feb 2022

- Developed NLP solutions for customer care applications and optimized Deep Learning models.
- Designed robust data pipelines and implemented end-to-end ML production workflows.

NLP RESEARCH SCOLARSHIP

University of Rome "Tor Vergata" in collaboration with AlmaWave

Rome, Italy | Sep 2020 - Feb 2021

- Conducted research on integrating rule injection in Neural Networks for dialog system development.
- Published "Every Time I Fire a Conversational Designer, the Performance of the Dialog System Goes Down" (LREC 2022).

SKILLS

- **Coding Languages:** Python, Java, Javascript
- **ML Frameworks:** Pytorch, Scikit-learn, Huggingface transformers, MarianMT, Pytorch Lightning
- **AI Agents Frameworks:** Langchain, Langgraph, Haystack, LlamaIndex
- **ML Inference:** ONNX, ONNXRuntime, NVIDIA TensorRT, NVIDIA Triton, vLLM
- **MLOps Tools:** Docker, Kubernetes, Git, MLFlow, Airflow, Github, Github Actions, DVC, AWS cloud services, SLURM, Prometheus, Grafana
- **Databases:** Postgres, MongoDB, Redis, MySQL

PUBBLICATIONS

COMBINING TRANSFORMER-BASED ARCHITECTURES AND TRANSFER LEARNING FOR ROBUST STANCE DETECTION

EVALITA | 2020
https://ceur-ws.org/Vol-2765/paper99.pdf

EVERY TIME I FIRE A CONVERSATIONAL DESIGNER, THE PERFORMANCE OF THE DIALOG SYSTEM GOES DOWN

LREC | 2022
https://aclanthology.org/2022.lrec-1.15.pdf

ADDITIONAL EXPERIENCE AND REWARDS

1ST PLACE AT EVALITA 2020 (ITALIAN NATIONAL NLP COMPETITION), SARDISTANCE TASK

Università degli studi di Roma "Tor Vergata"
Rome, Italy | Oct 2020

BEST SYSTEM AWARD

EVALITA 2020 (Italian National NLP Competition)
Dec 2020
https://www.evalita.it/campaigns/evalita-2020/best-system-award/

2ST PLACE AT HACKPARTY W21 (AI CHALLENGE)

Pi Campus
Rome, Italy | 2022

COMPUTER SCIENCE STUDENTS REPRESENTATIVE

Università degli studi di Roma "Tor Vergata"
Rome, Italy | 2017 - 2021

COMPUTER SCIENCE HACKATHON ORGANIZER

Università degli studi di Roma "Tor Vergata"
Rome, Italy | 2019

EDUCATION

MASTER DEGREE IN COMPUTER SCIENCE (110/110 CUM LAUDE)

University of Rome "Tor Vergata" | 2019 - 2021

Thesis: "Evaluation of Explainability of Machine Learning Models: ERO-MD Explainable Model for Migraine Diagnosis"
Supervisor: Prof. Fabio Massimo Zanzotto

BACHELOR'S DEGREE IN COMPUTER SCIENCE

University of Rome "Tor Vergata" | 2016-2019

Thesis: “Solution for automation in the semantic annotations and in the population of ontologies. the case of resolutions of the United Nations”
Supervisor: Prof. Armando Stellato

CERTIFICATIONS

- **DeepLearning.AI** – Introduction to Machine Learning in Production (2021)
- **Udemy** – Docker and Kubernetes: The Complete Guide (2021)
- **HackerRank** – REST API (Intermediate) (2020)

LANGUAGES

- **Italian** - Native language
- **English** - B2